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Impedance Identification Using Tactile Sensing and Its Adaptation for an Underactuated Gripper Manipulation

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Abstract: Underactuated gripper has a broad application in the field of space robot and industrial robot because of its better shape-adaptation. However, because of the underactuated characteristics, it is a great challenge to accurately obtain the displacement of the contact point between the finger and grasped object, which makes it difficult to control the gripper grasp stably, especially the environmental parameters are unknown. This paper develops the identification of the unknown environmental parameters using a tactile array sensor based on the recursive least-squares (RLS) method. The unknown environments are described as linear systems with unknown dynamics, and the environmental parameters are identified using the measured contact force and the derived displacement of the contact point which is obtained through the underactuated gripper dynamics. Meanwhile, an impedance adaptive control is presented to match the variability of the environment parameters, and the desired impedance model is imposed to the underactuated gripper to achieve stable grasp. A cost function that measures the contact force, velocity and displacement error is defined, and the critical impedance parameters are found to minimize it. At last, a co-simulation of ADAMS and MATLAB for an underactuated gripper grasp is implemented to show the feasibility of environmental parameters identification and its adaptive method.

Keywords: Environmental parameters identification, impedance adaptation, stable grasp, underactuated gripper.

1. INTRODUCTION

The end effectors play an important role in the field of space robot and industrial manipulator. As a type of end effectors, the underactuated hand has a broad application because of its better self-adaption, which can grasp various objects in diverse tasks as the fingers adapt themselves to the shape of the objects by their mechanical behavior [1]. Due to the underactuated characteristics, many different types of passive elements, such as springs, are considered to resolve the non-uniqueness question involved with null space grasp and ensure the shape-adaptation of the finger to the object grasp [2]. For the underactuated fingers, the appropriate application of the spring makes it possible for one input torque to drive a finger that has more than one DOF [3,4]. It is worth to mention that the balance between the contact forces, the input torque and the spring passive torque contributes to various grasp types and has a significant effect on stable grasp. For the underactuated fingers, the stable grasp is to control the input torque adapting the unknown environment. The contact force is associated with the properties of environment. Therefore, identification of the unknown environment parameters is the basis for stable grasp, while the input torque is appropriately controlled to guarantee the motion of gripper adapting the unknown environment.

In the aspect of identifying the unknown environment, there are many works have been done by researchers. Funchun Sun and Huaping Liu do many researches about the object recognition [5,6]. In [5], a dynamic algorithm is proposed to address T-S fuzzy system identification with both sparsity and dynamic clustering. [6] presents the tactile object classification based on tactile sequences, while the sequences are separated into series of subgroups and the features are extracted with the linear dynamical system (LDS). In [7], the mass, and stiffness matrices of mechanical system are calculated based on the Instrumented Variable (IV) method with measured forces and displacements. In [8], the measured contact force and the displacement of contact point between finger and object are used to be environment information. From previous researches [7, 8], the contact force and displacements are used to recognize or identify the unknown environment. However, because the underactuated gripper has more than one

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DOF, it is a great challenge to obtain the accurate displacement of the contact point, saying nothing of the environmental parameters identification. To the best of the author's knowledge, there are few studies about the environmental parameters identification with the underactuated gripper.

Furthermore, it is the key technology for stable grasp to control the input torque matching the variability of the environment parameters, while there is a balance between the input torque, spring passive torque and the contact force. In the literature of robot-environment interaction control, impedance control [9] has a broad application [10–12] because of its better robustness [13]. Under impedance control, the underactuated gripper's compliance is determined by the interaction force exerted by environment and the spring passive torque. Generally, a good adaptive grasp strategy is to build the connection between the desired impedance model [8] and the environment. As a result, impedance adaptation has been widely investigated. In [14], an impact controller based on the switching of the impedance parameters depending on dissipated and generated forms of energy is discussed. In [15], a novel policy iteration approach for finding online adaptive optimal controllers of linear systems with completely unknown system dynamics is proposed, while it is only for state regulation. In [8], the desired impedance model is obtained based on adaptive optimal control by using the trajectory of robot arm and the interaction force, but the robot's motion in the initial stage absolutely relies on the initial value of the impedance parameters.

Although impedance adaptation has been broadly developed, it is challenging to develop impedance control for underactuated gripper because the environmental parameters are unknown, and the identification of the environmental parameters usually requires the displacement of contact point and the contact force. However, it is difficult for the underactuated gripper to directly obtain the accurate displacement of contact point. To address this problem, this paper aims to obtain the displacement of contact point with the measured contact force for environmental parameters identification, based on which the impedance adaptation is developed to achieve stable grasp. Firstly, the dynamics of underactuated gripper is employed to obtain the displacement of contact point using a tactile array sensor. Then, the environmental parameters are identified based on the RLS method using the measured contact force and the derived displacement of the contact point. Finally, a desired impedance model is imposed on the underactuated gripper to guarantee a stable grasp, and the identified environmental parameters are used to adjust the desired impedance parameters based on the optimal control. The identification-control process will guarantee the desired impedance parameters match the environmental parameters.

The paper is organized as follows: Section 2 presents the description of the underactuated gripper grasping unknown object. In Section 3, the environmental parameters identification and impedance adaptive control are presented. In Section 4, the corresponding results are verified though co-simulation with ADAMS and MATLAB. Section 5 gives the conclusions.

2. THE DISCRIPTION OF UNDERACTUATED GRIPPER GRASPING UNKNOWN OBJECT

Fig. 1 presents the model of the underactuated gripper with an unknown environment. The unknown environment is described as mass-damping-stiffness system with unknown environmental parameters (contact mass, damping and stiffness) [8]. The underactuated gripper consists of 2 fingers with a rotational spring (installed between link L_1 and link L_2), which is in a different location from that of the SARAH hand [16] to resolve the indeterminacy so that distal phalanges can move relative to one another in the parallel manner with less energy consumption [17]. Each of the fingers is a closed-loop system with four moving linkages and a fixed linkage. T_a is the driving torque exerted by motor, and f is the contact force between the fingers and the environment. T_k is the passive torque exerted by the rotational spring. The φ_i ($i = 1, 2, 3, 4$) is the angle between linkage L_i and the X axes. The S denotes the position of contact point on the distal phalanx, and which is the distance along the y_1 axes from the contact point to the joint O_4. A tactile array sensor is fixed on the distal phalanx to measure the contact force and the position of contact point. The tactile array sensor is composed of several tactile units, and the contact point is at the position where the output of the tactile unit is maximum.

For a stable grasp, there is a balance between the input torque T_a, the passive spring torque T_k and the contact force f, which can be obtained from the post-grasp statics analysis. The key for stable grasp is to control the input torque enable the motion of underactuated fingers match the unknown environmental parameters. The post-grasp statics analysis and dynamics will be described in subsection 2.1 and subsection 2.2.

2.1. Statics analysis

For stable state, one obtains [2]

$$\mathbf{t}^{\mathbf{T}}\mathbf{w_a} = \mathbf{fv}, \tag{1}$$

where

$$\mathbf{t} = \begin{bmatrix} T_a \\ T_k \end{bmatrix}, \ \mathbf{f} = [f], \ \mathbf{v} = \left[v_{c_1}^{\mathrm{T}} x_1\right], \ \mathbf{w_a} = \begin{bmatrix} \dot{\varphi}_1 \\ \dot{\gamma} \end{bmatrix}, \tag{2}$$

where $\dot{\gamma}$ is the angular velocity between the linkages L_1 and L_2, $\mathbf{v}_{c_1}^{\mathrm{T}} x_1$ is the projected velocity of the contact point onto the x_1 axes, and $\dot{\varphi}_1$ is the angular velocity between linkage L_1 and X axes.

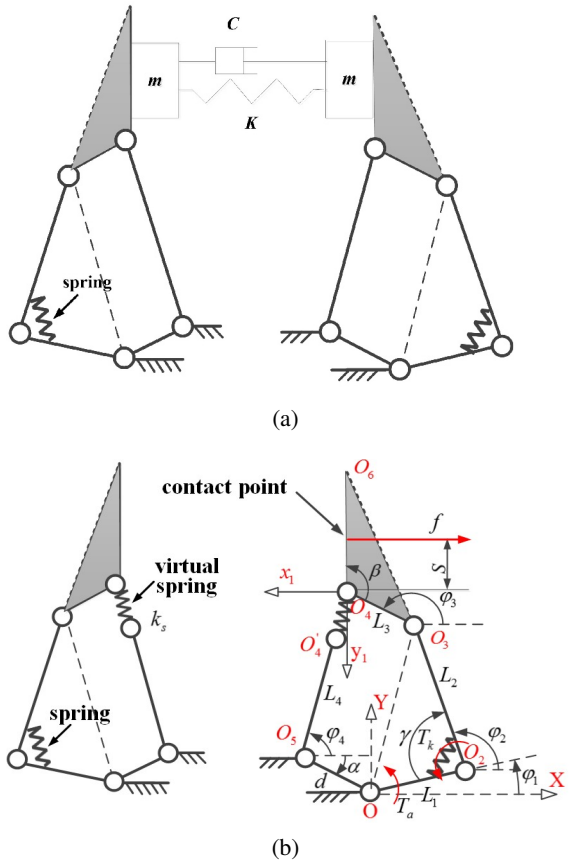


Fig. 1. 2-DOF underactuated gripper model with unknown environment.

According to the kinematics of the underactuated gripper (shown in Fig. 1), the projected velocity of the contact point can be expressed as the product of the Jacobian matrix $\mathbf{J}$ and the derivatives of the finger joints' coordinate:

$$\mathbf{v} = \left[v_{c_1}^{\mathrm{T}} x_1\right] = \mathbf{J}_0\mathbf{J}_1\mathbf{J}_2\mathbf{J}_3 \begin{bmatrix} \dot{\varphi}_1 \\ \dot{\gamma} \end{bmatrix}, \tag{3}$$

where

$$\mathbf{J}_0 = \begin{bmatrix} S + L_4\cos\theta & S \end{bmatrix}, \tag{4a}$$

$$\mathbf{J}_1 = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}, \tag{4b}$$

$$\mathbf{J}_2 = \begin{bmatrix} X_1 & Y_1 \\ X_2 & Y_2 \end{bmatrix} = \begin{bmatrix} -\frac{\partial f_4/\partial\varphi_1}{\partial f_4/\partial\varphi_4} & -\frac{\partial f_4/\partial\varphi_2}{\partial f_4/\partial\varphi_4} \\ -\frac{\partial f_3/\partial\varphi_1}{\partial f_3/\partial\varphi_3} & -\frac{\partial f_3/\partial\varphi_2}{\partial f_3/\partial\varphi_3} \end{bmatrix}, \tag{4c}$$

$$\mathbf{J}_3 = \begin{bmatrix} 1 & 0 \\ 1 & -1 \end{bmatrix}, \tag{4d}$$

where θ is the angle between line O_4O_6 and line O_5O_4, and

$$f_1 = d\cos\alpha + L_1\cos\varphi_1 + L_2\cos\varphi_2 + L_3\cos\varphi_3 - L_4\cos\varphi_4, \tag{5a}$$

$$f_2 = d\sin\alpha + L_1\sin\varphi_1 + L_2\sin\varphi_2 + L_3\sin\varphi_3 - L_4\sin\varphi_4, \tag{5b}$$

$$f_3 = d^2 + L_1^2 + L_2^2 + L_3^2 - L_4^2 + 2dL_1\cos(\alpha-\varphi_1) + 2dL_2\cos(\alpha-\varphi_2) + 2dL_3\cos(\alpha-\varphi_3) + 2L_1L_2\cos(\varphi_1-\varphi_2) + 2L_1L_3\cos(\varphi_1-\varphi_3) + 2L_2L_3\cos(\varphi_2-\varphi_3), \tag{5c}$$

$$f_4 = d^2 + L_1^2 + L_2^2 + L_4^2 - L_3^2 + 2dL_1\cos(\alpha-\varphi_1) + 2dL_2\cos(\alpha-\varphi_2) - 2dL_4\cos(\alpha-\varphi_4) + 2L_1L_2\cos(\varphi_1-\varphi_2) - 2L_1L_4\cos(\varphi_1-\varphi_4) - 2L_2L_4\cos(\varphi_2-\varphi_4). \tag{5d}$$

By substituting (2), (3) into (1), the statics equation can be written as

$$\mathbf{t}^{\mathrm{T}}\mathbf{w}_a = \mathbf{f}^{\mathrm{T}}\mathbf{v} = \mathbf{f}^{\mathrm{T}}\mathbf{J}_0\mathbf{J}_1\mathbf{J}_2\mathbf{J}_3\mathbf{w}_a. \tag{6}$$

Equation (6) can be written as

$$\mathbf{f} = (\mathbf{J}_0\mathbf{J}_0^{\mathrm{T}})^{-1}\mathbf{J}_0(\mathbf{J}_1^{\mathrm{T}})^{-1}(\mathbf{J}_2^{\mathrm{T}})^{-1}(\mathbf{J}_3^{\mathrm{T}})^{-1}t. \tag{7}$$

Furthermore, substituting (2), (3) into (7) and reformulating it, one can obtain [18]

$$f + \frac{Y_1}{S(X_1Y_2 - X_2Y_1)}T_a + \frac{X_1 + Y_1}{S(X_1Y_2 - X_2Y_1)}T_k = 0. \tag{8}$$

Equation (8) reflects the balance between T_a, T_k and f during the stable state. T_a is an active torque one can directly control. T_k is a passive parameter related to the joint angle:

$$T_k = k_r(\varphi_2 - \varphi_{20}), \tag{9}$$

where k_r is the stiffness coefficient of the rotational spring. The φ_{20} is the initial value of φ_2. The contact force f is related to the characteristics of the environment. Considering the unknown environment into mass-damping-stiffness system, the contact force can be described by the following equation (the detailed derivation can be seen in the appendix)

$$f = -m\ddot{x} - 2C\dot{x} - 2Kx, \tag{10}$$

where m, C and K are the unknown environmental parameters, which are related to the contact mass, damping and stiffness. $\ddot{x}$, $\dot{x}$and x are the acceleration, velocity and displacement of the contact point, which change along the x_1 axes.

2.2. Dynamics

The dynamic equation for the underactuated gripper can be described [18]:

$$\mathbf{M}(\boldsymbol{\varphi})\ddot{\boldsymbol{\varphi}} + \mathbf{C}(\dot{\boldsymbol{\varphi}}, \boldsymbol{\varphi}) + \mathbf{H}(\boldsymbol{\varphi}) = \boldsymbol{\tau}, \tag{11}$$

i.e.,

$$\begin{bmatrix} M_{11} & M_{12} & M_{13} & 0 \\ M_{12} & M_{22} & M_{23} & 0 \\ M_{13} & M_{23} & M_{33} & 0 \\ 0 & 0 & 0 & M_{44} \end{bmatrix} \begin{bmatrix} \ddot{\varphi}_1 \\ \ddot{\varphi}_2 \\ \ddot{\varphi}_3 \\ \ddot{\varphi}_4 \end{bmatrix} + \begin{bmatrix} C_1 \\ C_2 \\ C_3 \\ C_4 \end{bmatrix} + \begin{bmatrix} H_1 \\ H_2 \\ H_3 \\ H_4 \end{bmatrix} = \begin{bmatrix} \tau_1 \\ \tau_2 \\ \tau_3 \\ \tau_4 \end{bmatrix},$$

where $\mathbf{M}(\boldsymbol{\varphi})$ is the 4×4 inertia matrix of the finger, $C(\dot{\boldsymbol{\varphi}},\boldsymbol{\varphi})$ is the 4×4 vector that contains the centrifugal and Coriolis terms, $\mathbf{H}(\boldsymbol{\varphi})$ is the 4×1 vector of the elastic forces, and $\boldsymbol{\tau}$ is the 4×1 vector of the generalised forces, and

$$\begin{aligned}
M_{11} &= m_1L_1^2/3 + m_2L_1^2 + m_3L_1^2,\\
M_{12} &= (m_2L_2/2 + m_3L_2)L_1\cos(\varphi_1-\varphi_2),\\
M_{13} &= m_3L_1L_3\cos(\varphi_1-\varphi_3)/2,\\
M_{22} &= m_2L_2^2/3 + m_3L_2^2,\\
M_{23} &= m_3L_2L_3\cos(\varphi_2-\varphi_3)/2,\\
M_{33} &= m_3L_3^2/3,\\
C_1 &= \dot{\varphi}_2^2L_1(m_2L_2/2 + m_3L_2)\sin(\varphi_1-\varphi_2)\\
&\quad + \dot{\varphi}_3^2m_3L_1L_3\sin(\varphi_1-\varphi_3)/2,\\
C_2 &= -\dot{\varphi}_1^2L_1(m_2L_2/2 + m_3L_2)\sin(\varphi_1-\varphi_2)\\
&\quad + \dot{\varphi}_3^2m_3L_2L_3\sin(\varphi_2-\varphi_3)/2,\\
C_3 &= -\dot{\varphi}_2^2m_3L_2L_3\sin(\varphi_2-\varphi_3)/2\\
&\quad - \dot{\varphi}_1^2m_3L_1L_3\sin(\varphi_1-\varphi_3)/2,\\
C_4 &= 0,\\
H_1 &= -L_1k_s\Delta_x\sin\varphi_1 + L_1k_s\Delta_y\cos\varphi_1 + T_k,\\
H_2 &= -L_2k_s\Delta_x\sin\varphi_2 + L_2k_s\Delta_y\cos\varphi_2 - T_k,\\
H_3 &= -L_3k_s\Delta_x\sin\varphi_3 + L_3k_s\Delta_y\cos\varphi_3,\\
H_4 &= L_4k_s\Delta_x\sin\varphi_4 - L_4k_s\Delta_y\cos\varphi_4,\\
\tau_1 &= T_a - f_xL_1\sin\varphi_1 + f_yL_1\cos\varphi_1,\\
\tau_2 &= -f_xL_2\sin\varphi_2 + f_yL_2\cos\varphi_2,\\
\tau_3 &= -f_xL_3\sin\varphi_3 + f_yL_3\cos\varphi_3,\\
\tau_4 &= 0,
\end{aligned} \tag{12}$$

where m_i $(i=1,2,3)$ is the mass of linkage i, and k_s is the stiffness of virtual spring. f_x and f_y are the projections of f onto the x_1 and y_1 axes, respectively, and Δ_x and Δ_y are the projections of Δ onto the x_1 and y_1 axes, respectively, which can be described as:

$$\begin{aligned}
\Delta_x &= L_1\cos\varphi_1 + L_2\cos\varphi_2 + L_3\cos\varphi_3 - L_4\cos\varphi_4\\
&\quad - d\cos(\pi+\alpha),
\end{aligned} \tag{13a}$$

$$\begin{aligned}
\Delta_y &= L_1\sin\varphi_1 + L_2\sin\varphi_2 + L_3\sin\varphi_3 - L_4\sin\varphi_4\\
&\quad - d\sin(\pi+\alpha).
\end{aligned} \tag{13b}$$

The total torque applied to the underactuated finger can be divided into two parts:

$$\boldsymbol{\tau} = \mathbf{T} - \mathbf{T}_{\text{ext}}, \tag{14}$$

where $\mathbf{T}$ is the active motor torque of the joint, i.e., $\mathbf{T} = \begin{bmatrix} T_a & 0 & 0 & 0 \end{bmatrix}^{\mathrm{T}}$; $\mathbf{T}_{\text{ext}}$ is the reaction torque imposed by the external force, i.e., $\mathbf{T}_{\text{ext}} = \begin{bmatrix} -\tau_1' & -\tau_2 & -\tau_3 & -\tau_4 \end{bmatrix}^{\mathrm{T}}$, where $\boldsymbol{\tau}_1' = -f_xL_1\sin\varphi_1 + f_yL_1\cos\varphi_1$.

Thus the (11) can be rewritten as

$$\mathbf{T} = \mathbf{M}(\boldsymbol{\varphi})\ddot{\boldsymbol{\varphi}} + \mathbf{C}(\dot{\boldsymbol{\varphi}},\boldsymbol{\varphi}) + \mathbf{H}(\boldsymbol{\varphi}) + \mathbf{T}_{\text{ext}}. \tag{15}$$

A numerical method can be employed to obtain the joint angles $\boldsymbol{\varphi} = \begin{bmatrix} \varphi_1 & \varphi_2 & \varphi_3 & \varphi_4 \end{bmatrix}$ by solving (15) with the actuated torque T_a and the measured contact force f, such as Runge-Kutta method. Then, according to the geometrical relationship between the joint space and the work space (Fig. 2), the position of contact point in Cartesian space can be calculated:

$$\begin{aligned}
\mathbf{D} &= \begin{bmatrix} d_x & d_y \end{bmatrix}^{\mathrm{T}} - \begin{bmatrix} d_{x0} & d_{y0} \end{bmatrix}^{\mathrm{T}}\\
&= \begin{bmatrix} \cos\varphi_1 & \cos\varphi_2 & \cos\varphi_3 & \cos(\beta+\varphi_3-\pi) \\ \sin\varphi_1 & \sin\varphi_2 & \sin\varphi_3 & \sin(\beta+\varphi_3-\pi) \end{bmatrix} \begin{bmatrix} L_1 \\ L_2 \\ L_3 \\ S \end{bmatrix}\\
&\quad - \begin{bmatrix} \cos\varphi_{10} & \cos\varphi_{20} & \cos\varphi_{30} & \cos(\beta+\varphi_{30}-\pi) \\ \sin\varphi_{10} & \sin\varphi_{20} & \sin\varphi_{30} & \sin(\beta+\varphi_{30}-\pi) \end{bmatrix}\\
&\quad \times \begin{bmatrix} L_1 \\ L_2 \\ L_3 \\ S_0 \end{bmatrix},
\end{aligned} \tag{16}$$

where φ_{i0} is the initial value of φ_i; $\begin{bmatrix} d_{x0} & d_{y0} \end{bmatrix}^T$ is the initial position of the contact point, and $\begin{bmatrix} d_x & d_y \end{bmatrix}^T$ is the current position. Additionally, the position of contact point on the distal phalanx (variable S) can be measured by the tactile array sensor, i.e., where the output of the tactile unit is maximal.

Therefore, the displacement of contact point along the x_1 axes can be obtained from the geometrical relationship (seen in Fig. 2).

$$x = \|\mathbf{D}\|_2 \cdot \cos(\varphi_3 - \varphi_{30}). \tag{17}$$

Thus the displacement of contact point is obtained. As discussed in the introduction, it is the basis for developing the environmental parameters identification and impedance adaptation, which will be presented in the Section 3.

3. IMPEDANCE ADAPTIVE CONTROL

This section is dedicated to develop the environmental parameters identification and the impedance adaptive control-the corresponding scheme can be seen in Fig. 3, the detailed description can be found in the following.

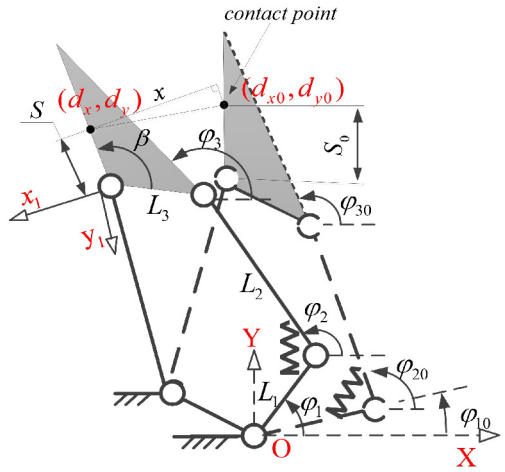


Fig. 2. Geometrical relationship between joint space and work space.

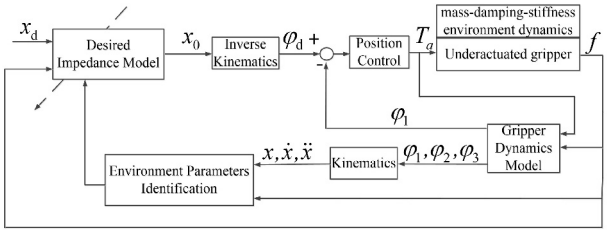


Fig. 3. Scheme of the environmental parameters identification and impedance adaptive control.

3.1. Preliminary: identification of the unknown environmental parameters based on RLS

Identification of the unknown environment parameters based on RLS is briefly introduced in this subsection, of which the results will be used for the development of the impedance adaptation.

Recalling (10), for each $k \in Z^+$ where k stands for kth step and Z^+ is positive integer set, the discretization of (10) is

$$f(k) = -m\ddot{x}(k) - 2C\dot{x}(k) - 2Kx(k). \tag{18}$$

Equation (10) can be rewritten as matrix form

$$\mathbf{f}(k) = -\mathbf{r}^{\mathrm{T}}(k)\mathbf{w}, \tag{19}$$

where

$$\mathbf{r}(k) = \begin{bmatrix} \ddot{x}(k) & 2\dot{x}(k) & 2x(k) \end{bmatrix}^{\mathrm{T}}, \tag{20}$$

$$\mathbf{w} = \begin{bmatrix} m & C & K \end{bmatrix}^{\mathrm{T}}. \tag{21}$$

The RLS estimation of $\mathbf{w}$ is [19],

$$\hat{\mathbf{w}}(t) = \hat{\mathbf{w}}(k-1) + \mathbf{L}(k)[\mathbf{f}(k) - \mathbf{r}^{\mathrm{T}}(k)\hat{\mathbf{w}}(k-1)], \tag{22}$$

$$\mathbf{L}(k) = \mathbf{P}(k-1)\mathbf{r}(k)[1 + \mathbf{r}^{\mathrm{T}}(k)\mathbf{P}(k-1)\mathbf{r}(k)]^{-1}, \tag{23}$$

$$\mathbf{P}(k) = [\mathbf{I}_3 - \mathbf{L}(k)r^{\mathrm{T}}(k)]\mathbf{P}(k-1), \tag{24}$$

where $\mathbf{P}(k)$ is the covariance matrix of dimension 3×3, and $\mathbf{P}(0) = p_0\mathbf{I}_3$ with p_0 is a large positive number, e.g., $p_0 = 10^6$. Defining the estimation error $\delta = \hat{\mathbf{w}}(k) - \mathbf{w}$, which approaches to zero when the input signal is persistently exciting and the noise has zero mean and finite variance [19]. Thus the environmental parameters are obtained with the RLS algorithm.

3.2. Impedance adaptive control

On the basis of the identification of the unknown environment parameters, the key step for a stable grasp is to control the input torque matching the variability of the environmental parameters. Then, the desired impedance model is defined as

$$f = -k_0\dot{x}_0 - k_1x_0 - k_2x_d, \tag{25}$$

where x_d is the desired trajectory of the contact point; x_0 is the virtual desired trajectory; k_0, k_1 and k_2 are impedance controller parameters to be determined. For an ideal position controller, it is obvious that $x_0(t) = x(t)$. Then the desired impedance model can be rewritten as

$$f = -k_0\dot{x} - k_1x - k_2x_d. \tag{26}$$

A state $\mathbf{z} \in \mathbf{R}^m$ is defined

$$\begin{cases} \dot{\mathbf{z}} = \mathbf{U}\mathbf{z}, \\ x_d = \mathbf{V}\mathbf{z}, \end{cases} \tag{27}$$

where $\mathbf{U} \in \mathbf{R}^{m\times m}$ and $\mathbf{V} \in \mathbf{R}^{n\times m}$ are two known matrices.

Formulating (10) and (27) together, one can obtain:

$$\begin{bmatrix} \ddot{x} \\ \dot{x} \\ \dot{\mathbf{z}} \end{bmatrix} = \begin{bmatrix} -2Cm^{-1} & -2Km^{-1} & \mathbf{0} \\ 1 & 0 & \mathbf{0} \\ 0 & 0 & \mathbf{U} \end{bmatrix} \begin{bmatrix} \dot{x} \\ x \\ \mathbf{z} \end{bmatrix} + \begin{bmatrix} -m^{-1} \\ 0 \\ \mathbf{0} \end{bmatrix} f. \tag{28}$$

Equation (28) can be rewritten into

$$\dot{\boldsymbol{\xi}} = \mathbf{A}\boldsymbol{\xi} + \mathbf{B}\mathbf{u}, \tag{29}$$

where $\boldsymbol{\xi} \in \mathbf{R}^n$ is the system state; $\mathbf{u} \in \mathbf{R}^m$ is the system input, and $\boldsymbol{\xi} = \begin{bmatrix} \dot{x} & x & \mathbf{z} \end{bmatrix}^{\mathrm{T}}$; $\mathbf{u} = [f]$. $\mathbf{A} \in \mathbf{R}^{n\times n}$ and $\mathbf{B} \in \mathbf{R}^{n\times m}$ are constant matrices, and

$$\mathbf{A} = \begin{bmatrix} -2Cm^{-1} & -2Km^{-1} & \mathbf{0} \\ 1 & 0 & \mathbf{0} \\ 0 & 0 & \mathbf{U} \end{bmatrix}, \tag{30}$$

$$\mathbf{B} = \begin{bmatrix} -m^{-1} & 0 & \mathbf{0} \end{bmatrix}^{\mathrm{T}}. \tag{31}$$

One will develop impedance adaptation such that the following cost function is minimized.

$$\boldsymbol{\Gamma} = \int_0^\infty (\dot{x}^{\mathrm{T}}Q_1\dot{x} + \begin{bmatrix} x^{\mathrm{T}} & x_d^{\mathrm{T}} \end{bmatrix} \begin{bmatrix} Q_2^2 & -Q_2^2 \\ -Q_2^2 & Q_2^2 \end{bmatrix} \begin{bmatrix} x \\ x_d \end{bmatrix}$$

$$
\begin{aligned}
&+ f^{\mathrm{T}}\mathbf{R}f)dt \\
&= \int_0^{\infty} (\dot{x}^{\mathrm{T}} Q_1 \dot{x} + \begin{bmatrix} x^{\mathrm{T}} & \mathbf{z}^{\mathrm{T}} \end{bmatrix} \begin{bmatrix} Q_2^2 & -Q_2^2\mathbf{V} \\ -\mathbf{V}^{\mathrm{T}}Q_2^2 & \mathbf{V}^{\mathrm{T}}Q_2^2\mathbf{V} \end{bmatrix} \begin{bmatrix} x \\ \mathbf{z} \end{bmatrix} \\
&+ f^{\mathrm{T}}\mathbf{R}f)dt \\
&= \int_0^{\infty} [\xi^{\mathrm{T}} Q \xi + f^{\mathrm{T}}\mathbf{R}f]dt. \qquad (32)
\end{aligned}
$$

The cost function presented in (32) represents the compromise/combination of the force regulation and trajectory tracking and determine the interaction performance. In (32), $\mathbf{Q}$ and $\mathbf{R}$ are the weighting matrices which satisfy

$$
\mathbf{Q} = \mathbf{Q}^{\mathrm{T}} = \begin{bmatrix} Q_1^2 & 0 & 0 \\ 0 & Q_2^2 & -Q_2^2\mathbf{V} \\ 0 & -\mathbf{V}^{\mathrm{T}}Q_2^2 & \mathbf{V}^{\mathrm{T}}Q_2^2\mathbf{V} \end{bmatrix}, \quad \mathbf{R} = [r]. \qquad (33)
$$

The factors Q_1 and Q_2 satisfy: $Q_1 > 0$, $Q_2 > 0$, and $r > 0$. For certain $\mathbf{Q}$ and $\mathbf{R}$, solving the following Riccati equation

$$
\mathbf{PA} + \mathbf{A}^{\mathrm{T}}\mathbf{P} - \mathbf{PBR}^{-1}\mathbf{B}^{\mathrm{T}}\mathbf{P} + \mathbf{Q} = \mathbf{0}. \qquad (34)
$$

The optimal feedback gain matrix can be obtained

$$
\mathbf{K}_{\mathrm{k}} = -\mathbf{R}^{-1}\mathbf{B}^{\mathrm{T}}\mathbf{P}. \qquad (35)
$$

Thus, the desired impedance model has been obtained according to $\mathbf{K}_{\mathrm{k}}$ and (26),

$$
f = -\mathbf{K}_{\mathrm{k}}\boldsymbol{\xi} = -k_0\dot{x} - k_1 x - k_2 x_d. \qquad (36)
$$

As presented in Fig. 3, the virtual desired trajectory x is obtained according to (36) with measured f and given x_d, and the inner-position controller is to guarantee the trajectory tracking.

In this way, the stable grasp is achieved and (36) is the resulted critical impedance function in the presence of variable environment dynamics.

3.3. Discussion

For post grasp configuration, it is obvious that $\ddot{x}$ and $\dot{x}$ are equal to zeroes at final stage. i.e., $\ddot{x} = 0$ and $\dot{x} = 0$. Substituting this condition into (10) and (36) respectively lead to

$$
f = -2Kx, \qquad (37)
$$

$$
f = -k_1 x - k_2 x_d. \qquad (38)
$$

Substituting (37) into (38), the stable displacement is obtained:

$$
x = \frac{-k_2}{k_1 - 2K} x_d. \qquad (39)
$$

Therefore, the final contact force is obtained:

$$
f = -2Kx = \frac{2Kk_2}{2K - k_1} x_d, \qquad (40)
$$

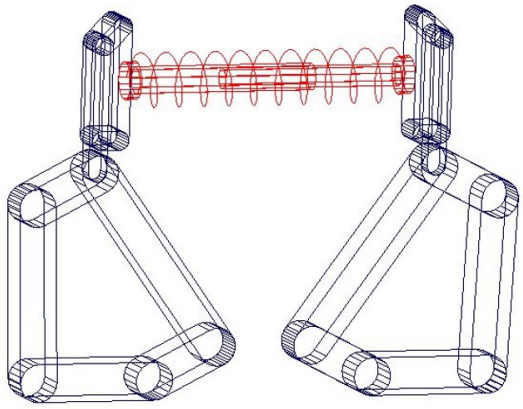


Fig. 4. Adams model of the underactuated finger with mass-stiffness-damping environment.

where the k_1 and k_2 can be obtained from Riccati equation

$$
k_1 = -\sqrt{Q_2^2/r + 4K^2} + 2K, \qquad (41)
$$

$$
\begin{aligned}
k_2 = {} & \mathbf{Q}_2^2\mathbf{V}r^{-1}(k_0\mathbf{U} - 2\mathbf{CU} + m\mathbf{U}^2 + 2K\mathbf{I} - k_1\mathbf{I})^{-1} \\
& \times \mathbf{V}^{\mathrm{T}}(\mathbf{V}\mathbf{V}^{\mathrm{T}})^{-1}, \qquad (42)
\end{aligned}
$$

where

$$
k_0 = -\sqrt{Q_1^2/r - 2mk_1 + 4C^2} + 2C. \qquad (43)
$$

From (39) to (43), it can be seen that the stable displacement and contact force depend on the following parameters:

I: Environment parameters: contact mass, damping and stiffness (m, C, K);

II: The weighting parameters: $\mathbf{Q}$, $\mathbf{R}$, $\mathbf{U}$ and $\mathbf{V}$;

III: The desired trajectory: x_d.

4. SIMULATION STUDY

4.1. Simulation conditions

In this section, a model based on the co-simulation of ADAMS and MATLAB is introduced, as shown in Fig. 4, and the interface scheme is shown in Fig. 5. Specification of the underactuated gripper is shown in Table 1, and the φ_{i0} $(i = 1, 2, 3, 4)$ is the initial value of φ_i. Considering the diversity of environment, parameters with different order of magnitude has been defined in Table 2 to represent soft, medium and hard objects respectively. The model 1 with small stiffness presents soft objects. The model 2 with middle stiffness presents middle-hard objects. The model 3 with large stiffness presents hard objects. The position control is to guarantee the trajectory tracking in joint space, which is achieved by PID controller.

The weighting matrices are chosen as: $\mathbf{Q}_1 = 1$, $\mathbf{Q}_2 = 1500$ and $\mathbf{R} = [1]$. The parameters $\mathbf{U}$ and $\mathbf{V}$ are chosen

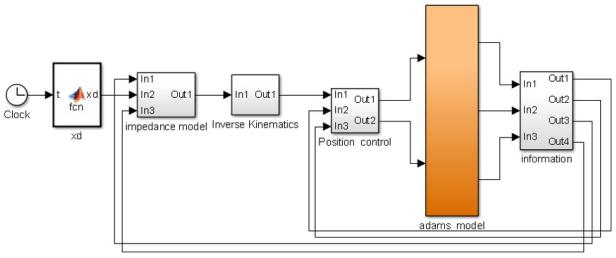


Fig. 5. Interface scheme of ADAMS and MATLAB.

Table 1. Parameters of the underactuated gripper.

Symbol	Value	Symbol	Value
L_1	0.0318 m	φ_{10}	$-4.482°$
L_2	0.0462 m	φ_{20}	$86.547°$
L_3	0.0190 m	φ_{30}	$153.430°$
L_4	0.0571 m	φ_{40}	$52.710°$
m_1	0.1 kg	d	0.0190 m
m_2	0.1 kg	k_s	1.0×10^6 N/m
m_3	0.2 kg	α	$-26.570°$
m_4	0.1 kg	β	$120.000°$
k_r	1 N/rad		

Table 2. Environment parameters.

	Mass (kg)	Damping (Ns/m)	Stiffness (N/m)
Model1	$m = 0.1$	$C = 1.0$	$K = 100$
Model2	$m = 0.5$	$C = 2.0$	$K = 500$
Model3	$m = 1.0$	$C = 0.5$	$K = 1000$

as: $\mathbf{U} = \begin{bmatrix} 0 & -1 \\ 0 & 0.5 \end{bmatrix}$ and $\mathbf{V} = \begin{bmatrix} 1 & 0.01 \end{bmatrix}$. And the desired trajectory is chosen as $x_d = 0.01 - 0.01e^{-0.5t}$, thus $\mathbf{Z} = \begin{bmatrix} 0.01 & -e^{-0.5t} \end{bmatrix}^{\mathrm{T}}$. Employing $f = -\mathbf{K}_0\boldsymbol{\xi} + \nu$ as the impedance model on the time interval [0s, 3s], where $\mathbf{K}_0$ is the initial value of the impedance parameters and ν is the exploration noise to satisfy the persistent excitation (PE) condition [8]. In the simulation, the $\mathbf{K}_0$ is set as $\mathbf{K}_0 = \begin{bmatrix} -1 & -1000 & 1000 \end{bmatrix}$ and the exploration noise is chosen as $\nu = \sum_{w=1}^{10} \frac{4}{w}\sin(wt)$. Therefore, the impedance model for this time is $f = \dot{x} + 1000x - 1000x_d + \nu$. The environmental parameters are identified at the whole process, at the same time, the impedance parameters are calculated. The impedance parameters are used to update the impedance model over the time interval (3s, 20s). The impedance model in this time period is the desired impedance model $f = -\mathbf{K}_k\boldsymbol{\xi}$.

4.2. Impedance adaptation

In this subsection, the three sets of environments presented in Table 2 are studied. The simulation results are shown in Fig. 6 and Fig. 7. In Fig. 6, the displacement

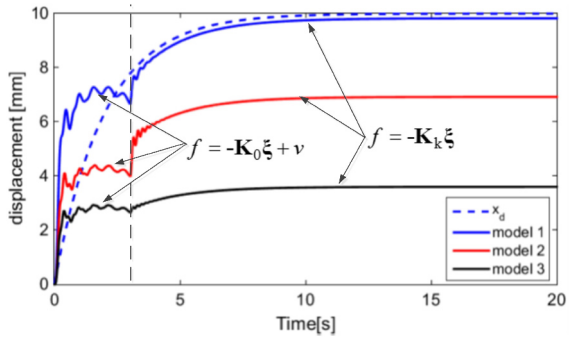


Fig. 6. Desired trajectory and actual displacement.

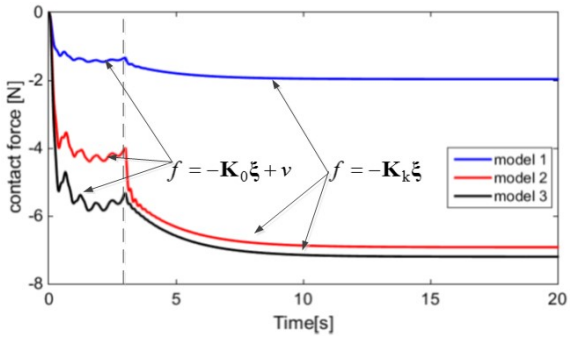


Fig. 7. Contact force.

of contact point in $x_1O_4y_1$ coordinate system is shown. In Fig. 7, the contact force between the underactuated finger and environment is shown.

In the first three seconds, there is a big oscillation on displacement and contact force. The big oscillation is caused by the exploration noise ν which is added to the impedance model to accelerate the estimation of environment parameters. For the three sets of environments, the displacement and contact force converge to constants. And for different models, the stable displacements and contact forces are different. The stable value of displacement and contact force depend on the impedance parameters. The convergence performance of impedance parameters are shown in Figs. 8-10.

For a certain environment, the impedance parameters converge to fixed values, based on which the desired impedance models are obtained with the proposed method. The desired impedance models of the three models are presented in Table 3 and compared to the ones calculated based on the nominal values of the contact mass, damping and stiffness. It is found that the obtained impedance models are very near to the ones under the nominal values. That is due to the identified results of the environmental parameters, which are shown in Figs. 11-13. The mean values of the identified results are presented in Table 5 and compared with nominal values.

To further investigate the effect of noise on the impedance adaptation, a white noise (standard deviation

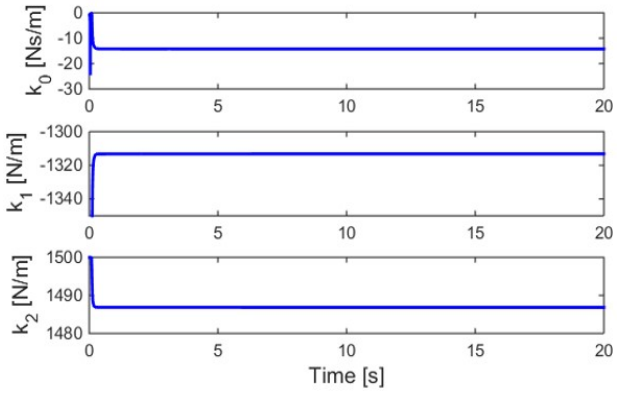


Fig. 8. Impedance parameters of Model 1.

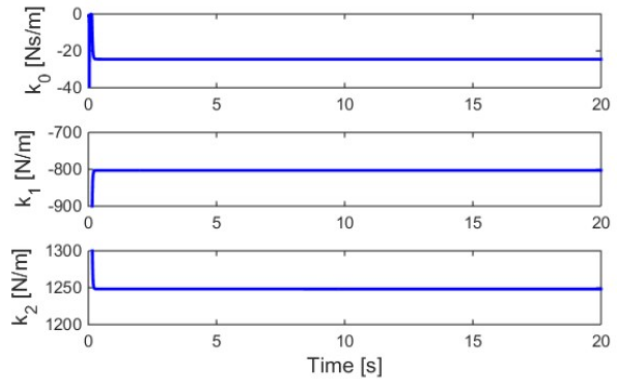


Fig. 9. Impedance parameters of Model 2.

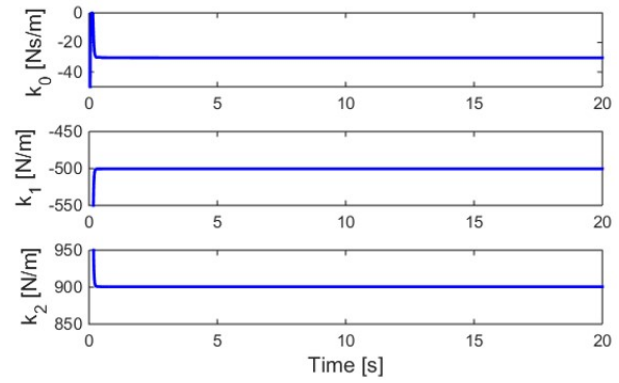


Fig. 10. Impedance parameters of Model 3.

is 0.1 N) is added to the contact force to simulate the real sensor. The simulation results are shown in Figs. 14 and 15. It is found that the displacements are very near to the ones without noise although there is a noise in contact force. At the condition of including the noise effect, the convergence performance of impedance parameters are shown in Figs. 16-18. The desired impedance models are presented in Table 4. The identified results of environmental parameters are shown in Figs. 19-21, and the mean values in 6 s-20 s are listed in Table 5.

Table 3. Impedance model.

	Desired impedance model	Impedance model based on nominal value
Model1	$f = 14.36\dot{x} + 1313.27x - 1486.83x_d$	$f = 14.23\dot{x} + 1313.28x - 1486.84x_d$
Model2	$f = 24.63\dot{x} + 802.76x - 1248.00x_d$	$f = 24.55\dot{x} + 802.95x - 1248.16x_d$
Model3	$f = 30.65\dot{x} + 500.00x - 899.98x_d$	$f = 30.37\dot{x} + 500.34x - 900.46x_d$

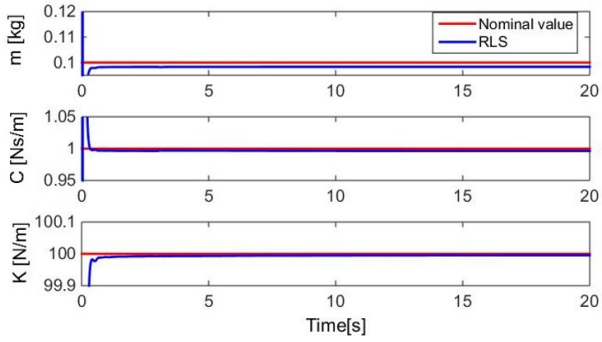


Fig. 11. Identified results of Model1 without noise.

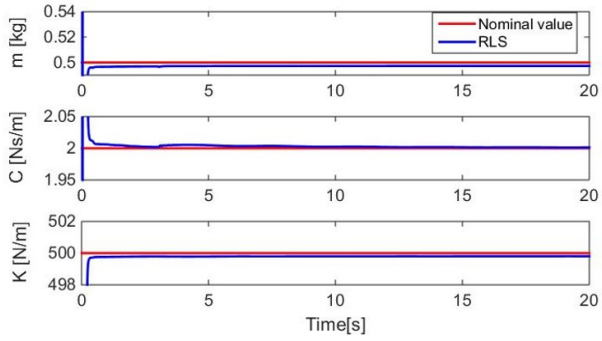


Fig. 12. Identified results of Model2 without noise.

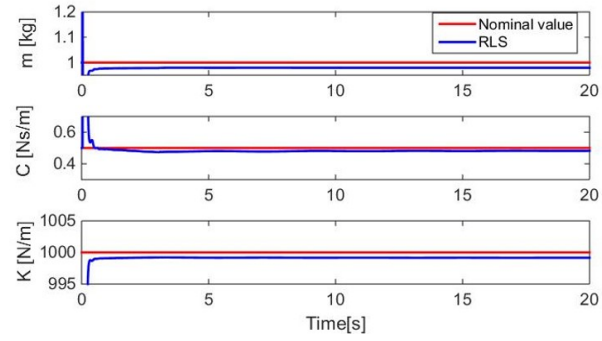


Fig. 13. Identified results of Model3 without noise.

In summary, it has been shown that the proposed impedance model can guarantee the stability of grasp, and the displacement and contact force are adjusted to adapt the different environments. The impedance models with

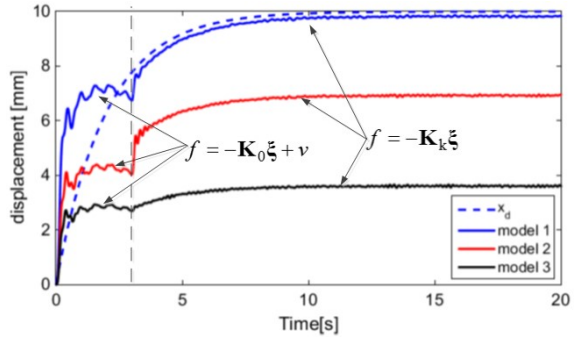


Fig. 14. Desired trajectory and actual displacement with noise.

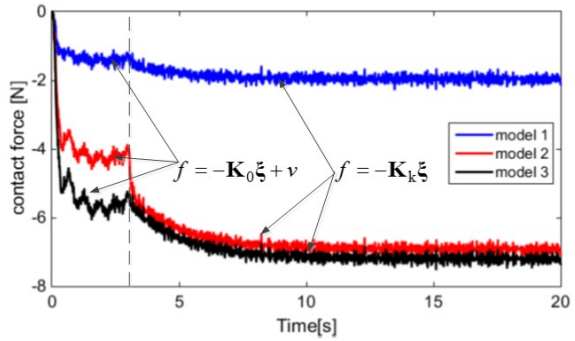


Fig. 15. Contact force with noise.

Table 4. Desired impedance models with noise.

	Desired impedance model
Model 1	$f = 14.54\dot{x} + 1313.57x - 1486.76x_d$
Model 2	$f = 24.18\dot{x} + 803.15x - 1248.34x_d$
Model 3	$f = 30.22\dot{x} + 500.55x - 900.77x_d$

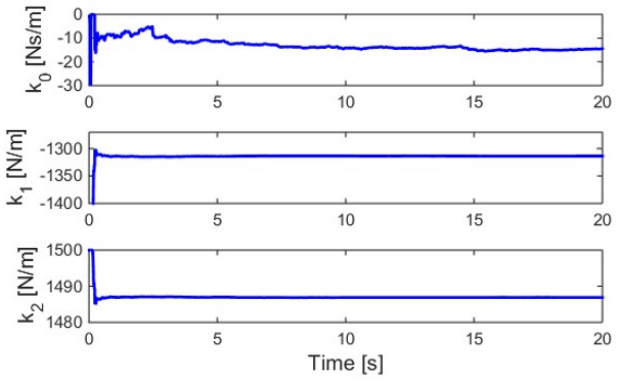


Fig. 16. . Impedance parameters of Model 1.

noise (shown in Table 4) are very close to the models without noise (shown in Table 3), although the damping has a bad identified result with noise. Possibly the coefficient of damping presented in (41)-(43) is too small compared to

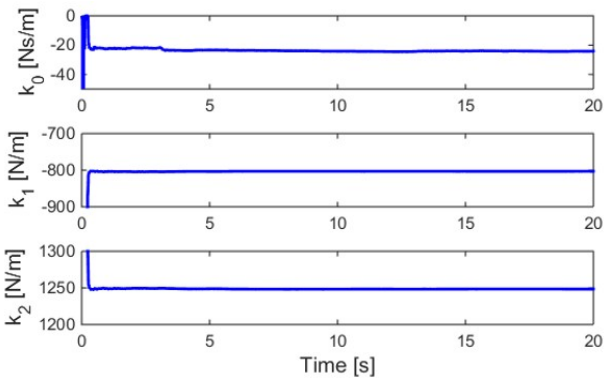


Fig. 17. Impedance parameters of Model 2.

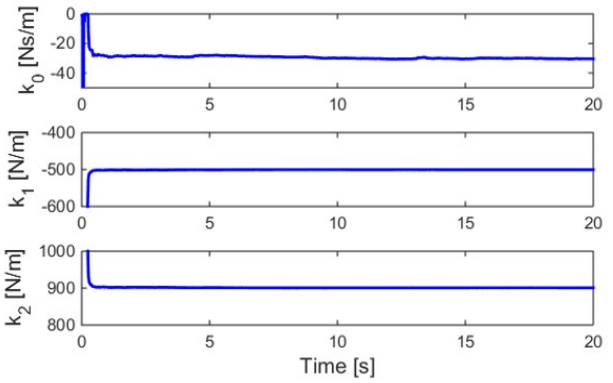


Fig. 18. Impedance parameters of Model 3.

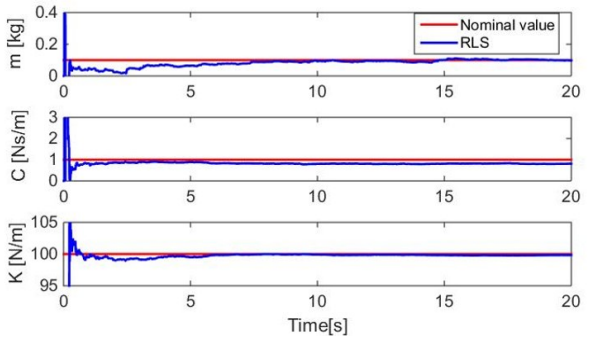


Fig. 19. Identified results of Model1 with noise.

Table 5. Identified results.

	Nominal value	RLS (without noise)	RLS (with noise)
Model 1	$m = 0.10$ Kg	0.0983	0.0980
	$C = 1.0$ Ns/m	0.9963	0.8108
	$K = 100$ N/m	99.9952	99.8326
Model 2	$m = 0.5$ Kg	0.4972	0.4722
	$C = 2$ Ns/m	2.0017	1.8102
	$K = 500$ N/m	499.8012	499.5750
Model 3	$m = 1.0$ Kg	0.9793	0.9665
	$C = 0.5$ Ns/m	0.4811	0.4594
	$K = 1000$ N/m	999.1586	998.6241

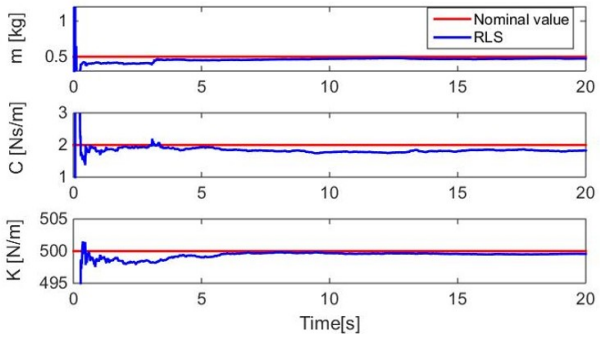


Fig. 20. Identified results of Model2 with noise.

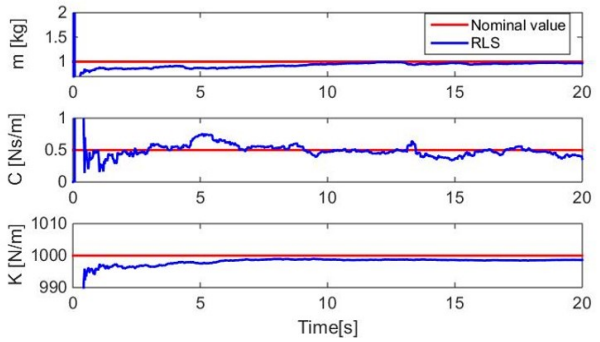
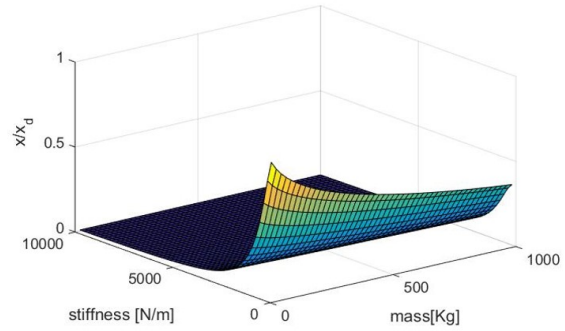
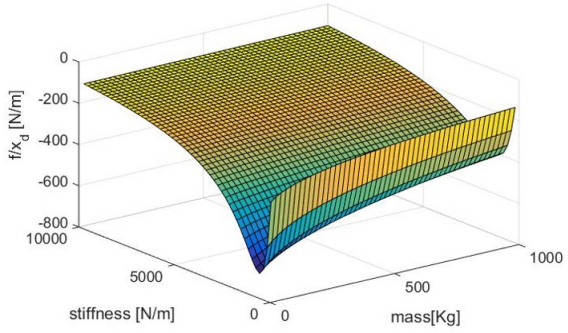


Fig. 21. Identified results of Model3 with noise.



(a)



(b)

Fig. 22. (a) The relationship between stable displacement and m/K; (b) the relationship between stable force and m/K.

the coefficients of contact mass and stiffness, so the damping has a little effect on the results.

4.3. Discussion

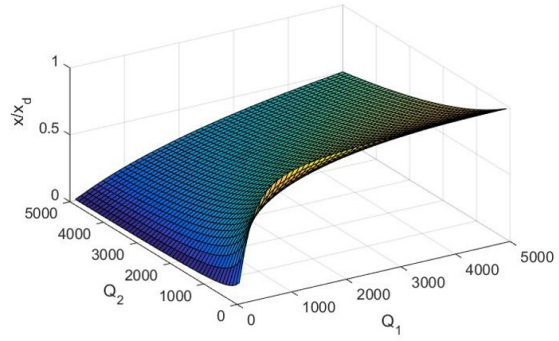
The environmental parameters, weighting parameters and desired trajectory will affect the stable displacement and contact force. Considering the coefficient of all parameters presented in (41)-(43), the contact mass, stiffness, and the weighting parameters Q_1 and Q_2 are just discussed in this paper. Fig. 22 shows the relationship between the stable displacement/force and m/K, where Q_1, Q_2, **U**, **V** and **R** are chosen as previous, while C is set as 1.0 Ns/m. Fig. 23 presents the relationship between the stable displacement/force and Q_1/Q_2, where **U**, **V** and **R** are chosen as before, and environmental parameters are chosen as same as model 1's.

Generally, the desired trajectory is chosen within the work space, while **Q** and **R** are chosen considering many factors, such as the minimum energy, displacement of contact point or the value of contact force. Once the Q_1 and Q_2 are selected, the parameters are just suitable for a part of environments. From Fig. 22(b), it can be found that the contact force approaches to zero with too large stiffness or too small stiffness, which may cause the instability of grasp. For the environments with large stiffness, bigger Q_1 and Q_2 may be suitable. On the contrary, for environment with small stiffness, smaller Q_1 and Q_2 may be suitable. Moreover, for certain Q_1 and Q_2, there is a biggest force (as shown in Fig. 23(b)), and one must guarantee the safety of the object in this condition. If the force is too big, proper Q_1 and Q_2 can be chosen to avoid this problem, these will give one a significant supervise for selecting suitable weighting matrices such as Q_1, Q_2 in the further research work.

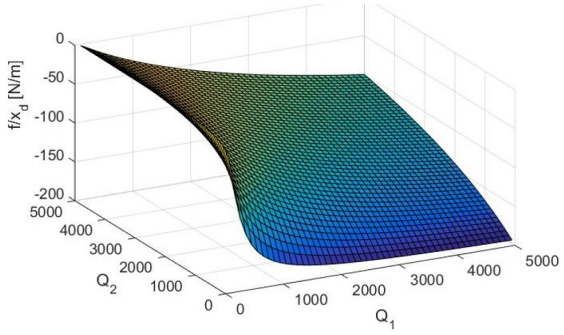
5. CONCLUSION

In this paper, the impedance adaptation has been investigated to achieve stable grasp for an underactuated gripper interacting with an unknown environment, based on which the desired impedance model has been obtained to match the unknown environment. The simulation shows that the proposed impedance model can guarantee the stability of grasp, and also present adaptation for different environments. Moreover, the stable displacement and contact force rely on the environmental parameters, weighting parameters and the desired trajectory.

Further works will focus on the experimental validation. A system including a two-finger underactuated gripper with DC motor, encoder and PPS (Pressure Profile System) tactile sensors has been established in the lab to



(a)



(b)

Fig. 23. (a)The relationship between stable displacement and Q_1/Q_2; (b) the relationship between stable force and Q_1/Q_2.

verify the validation of the proposed identified method and control method. In the hardware experiment, various compressive springs can be selected as the unknown environments to be grasped. The contact force can be measured by the PPS tactile sensors, and the rotation angle of DC motor can be obtained by the encoder. DSP is used to guarantee the data collection, data handling and controlling the motion of the underactuated gripper.

APPENDIX A

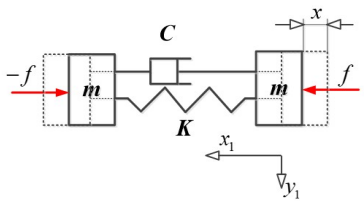
A.1. Proof of Equation (10)

Considering symmetric grasping, the mass-damping-stiffness environment contact with the two fingers at same time. The environment would undergo deformation due the force exerted by the fingers, as shown in Fig. 24(a). f_u is the force exerted by the finger, and

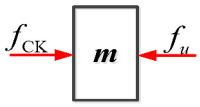
$$f_u = -f. \tag{A.1}$$

Considering the mass model, it is contact with finger and damping-stiffness model. Fig. 24(b) shows the force on mass model, where f_{CK} is the force exerted by damping-stiffness model. One obtains

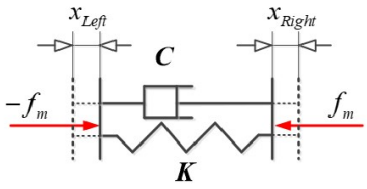
$$f_u - f_{CK} = m\ddot{x}. \tag{A.2}$$



(a)



(b)



(c)

Fig. 24. Force analysis.

Analyzing the damping-stiffness model, it contacts with mass model at left side and right side. Therefore, the model would undergo deformation at left side and right side, as shown in Fig. 24(c). Thus the force between damping-stiffness model and mass model can be obtained:

$$f_m = C(-\dot{x}_{Left} + \dot{x}_{Right}) + K(-x_{Left} + x_{Right}), \tag{A.3}$$

where x_{Left} is the deformation at left side and x_{Right} is the deformation at right side; $\dot{x}_{Left}$ and $\dot{x}_{Right}$ are the deformation rate of left side and right side, which satisfy $-x_{Left} = x_{Right} = x$, $-\dot{x}_{Left} = \dot{x}_{Right} = \dot{x}$. Thus (46) can be rewritten to

$$f_m = 2C\dot{x} + 2Kx. \tag{A.4}$$

Considering the interaction, one obtains

$$f_m = f_{CK}. \tag{A.5}$$

Substituting (47) and (48) into (45) leads to

$$f_u = m\ddot{x} + 2C\dot{x} + 2Kx. \tag{A.6}$$

Substituting (44) into (49), one can obtain

$$f = -m\ddot{x} - 2C\dot{x} - 2Kx. \tag{A.7}$$

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